

# Mingxuan Liu

Center of Computational Quantitative Medicine  
Duke-NUS Medical School

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## EDUCATION

### Duke-NUS Medical School

2022-Present

PhD in Biostatistics & Health Data Science, *Advisor: Nan Liu*

Exchange:

Mar 2024-Jul 2024

Boston Children's Hospital, Harvard Medical School, *Advisor: William La Cava*

### National University of Singapore

2020-2021

MS in Statistics (4.85/5, top 10%)

### Nanjing University

2016-2020

BSc in Mathematics, *Advisor: Libong Wang*

Exchange: University of Queensland

Aug 2016 - Dec 2016

## RESEARCH AREAS

AI fairness in healthcare, interpretable machine learning, biostatistics, missing value imputation

## PUBLICATIONS

- 2025     Zhu, M.\*, **Liu, M.\***, Ning, Y., Han, Y., Luo, Z., and Liu, N.  
Attention-Guided Fair AI Modeling for Skin Cancer Diagnosis. ***In preparation.***
- 2025     **Liu, M.**, Ning, Y., Wang, H., Hong, C., Engelhard, M., Bitterman, D.S.B., La Cava, W.G.,  
and Liu, N.  
Equitable Survival Prediction: A Fairness-Aware Survival Modeling (FASM) Approach.  
***Under review*** doi.org/10.48550/arXiv.2510.20629.
- 2025     **Liu, M.**, Ke, Y., Zhu, W., Mertens, M., Ning, Y., Liao, J., Hong, C., Ting, D.S.W., Peng, Y.,  
Bitterman, D.S., Ong, M.E.H. and Liu, N.  
Gender Bias in Large Language Models for Healthcare: Assignment Consistency and Clinical  
Implications. ***Under review*** doi.org/10.48550/arXiv.2510.08614.
- 2025     **Liu, M.\***, Ning, Y.\*, Teixayavong, S., Liu, X., Mertens, M., ... and Liu, N.  
A scoping review and evidence gap analysis of clinical AI fairness. ***Npj digital medicine***  
doi.org/10.1038/s41746-025-01667-2.
- 2024     **Liu, M.**, Ning, Y., Ke, Y., Shang, Y., Chakraborty, B., Ong, M.E.H., Vaughan, R. and Liu, N.  
Fairness-Aware Interpretable Modeling (FAIM) for Trustworthy Machine Learning in  
Healthcare. ***Patterns*** doi.org/10.1016/j.patter.2024.101059.

- 2023 **Liu, M.\***, Ning, Y.\*, Teixayavong, S., Mertens, M., Xu, J., ... and Liu, N.  
Towards clinical AI fairness: A translational perspective. *Npj digital medicine*  
doi:10.1038/s41746-023-00918-4.
- 2023 Xie, F., Ning, Y., **Liu, M.**, Li, S., Saffari, S. E., Yuan, H., Volovici, V., Ting, D. S. W., Goldstein, B. A., Ong, M. E. H., Vaughan, R., Chakraborty, B., and Liu, N.  
A universal AutoScore framework to develop interpretable scoring systems for predicting common types of clinical outcomes. *STAR protocols* doi:10.1016/j.xpro.2023.102302.
- 2023 **Liu, M.\***, Li, S.\*, Yuan, H., Ong, M.E.H., Ning, Y., Xie, F., Saffari, S.E., Shang, Y., Volovici, V., Chakraborty, B. and Liu, N.  
Handling missing values in healthcare data: A systematic review of deep learning-based imputation techniques. *Artificial Intelligence in Medicine*.  
doi:10.1016/j.artmed.2023.102587.
- 2022 Liu, N., **Liu, M.**, Chen, X., Ning, Y., Lee, J. W., Siddiqui, F. J., Saffari, S. E., Ho, A. F. W., Shin, S. D., Ma, M. H., Tanaka, H., Ong, M. E. H., and PAROS Clinical Research Network Investigators.  
Development and validation of an interpretable prehospital return of spontaneous circulation (P-ROSC) score for patients with out-of-hospital cardiac arrest using machine learning: A retrospective study. *EClinicalMedicine*. doi:10.1016/j.eclinm.2022.101422.

## CONFERENCE ABSTRACTS (PEER-REVIEWED)

- 2025 Ning, Y.\*, **Liu, M.**, Zhu, M., Ma, Y., Li, X., and Liu, N.  
Variable importance cloud for medical image analysis. In *AMIA Annual Symposium*. [Poster]
- 2024 **Liu, M.**, Ning, Y., Ke, Y., Shang, Y., Chakraborty, B., Ong, M.E.H., Vaughan, R. and Liu, N.  
Fairness-Aware Interpretable Modeling (FAIM) for Trustworthy Machine Learning in Healthcare. In *Symposium on Artificial Intelligence in Learning Health Systems (SAIL)*. [Poster]
- 2024 Zhu, M., **Liu, M.**, Yuan, H. and Liu, N.  
Attention-Guided Fair AI Modelling for Skin Cancer Diagnosis. In *Symposium on Artificial Intelligence in Learning Health Systems (SAIL)*. [Poster]
- 2024 **Liu, M.**, Ning, Y., Ke, Y., Shang, Y., Chakraborty, B., Ong, M.E.H., Vaughan, R. and Liu, N.  
Fairness-Aware Interpretable Modeling (FAIM) for Trustworthy Machine Learning in Healthcare. In *AMIA Informatics Summit*. [Poster]
- 2023 **Liu, M.**, Ning, Y., Yau, C.E., Okada, Y., Li, S., Ho, A.F.W., Shin, S.D., Ma, M.H.M., Tanaka, H., Ong, M.E.H., and Liu, N.  
Federated transfer learning for prediction of 30-day survival for out-of-hospital cardiac arrest patients. In *EMS Asia*. [Oral]

## AWARDS AND HONORS

- Khoo Pre-Doctoral Fellowship, Duke-NUS Medical School, 2022-2026

- Travel Support Award, Conference on Health, Inference, and Learning (CHIL), 2024
- Travel Support Award, Symposium on Artificial Intelligence in Learning Health Systems (SAIL), 2024
- Undergraduate Research Exchange Awards, Nanjing University, 2019
- The second prize of the Contemporary Undergraduate Mathematical Contest in Modelling, 2018
- Merit Student (Top 10% Students), Nanjing University, 2017



## **PROFESSIONAL EXPERIENCE**

### **Data Science Manager Intern**

American Express (supervised by Dr. Zhang Li), 2025

- Led the investigation of semi-supervised fine-tuning (SSF) strategies for both transformer encoder-only (e.g., ModernBART) and decoder-only (e.g., Gemma-2) architectures on natural language inference (NLI) tasks, spearheading the development and writing of a forthcoming manuscript.
- Applied SSF to detect factual and causal errors in large language model (LLM) explanations, achieving over 50% performance gains under low-labeling-budget settings and marginal improvements under high-labeling-budget scenarios. The results were presented to multiple technical and cross-functional audiences within the company.

### **Research Assistant Intern**

A\*star – Agency for Science, Technology and Research (supervised by Dr. Chin Su), 2021

- Conducted machine learning models (e.g., Random Forest and XGboost) via R and python to predict protein production based on frequency of amino acids, improving the general performance by 12% and writing pipelines to make the procedure efficient.

### **Data Science Intern**

Benshi.ai, 2020-2021

- Analyzed in project about maternal health and dealt with big datasets via Python and Spark through Azure cloud platform, providing efficient data analysis pipeline for modelling.
- Applied survival models via Pysurvival module to predict users' active time for Safe Delivery App and explored users' profiles, generating data insights with team members.
- Managed corporation cross teams with development of Machine Learning models and promote friendly cooperation with Denmark company Maternity Foundation.

## **CERTIFICATES**

- Deep Learning Specialization (5 courses) from Deeplearning.ai and Coursera
- Big Data Fundamentals with Pyspark from DataCamp
- Finding Hidden Messages in DNA (with Honors) from the University of San Diego and Coursera
- Design and Interpretation of Clinical Trials from Johns Hopkins University and Coursera
- Introduction to Healthcare from Stanford University and Coursera

Updated November 2025